Breadth-First Search for Shortest Paths

Topic: CompetitiveProgramming File: c5.pdf

PROBLEM

Find the minimum number of edges from a source vertex to all reachable vertices in an unweighted graph.

APPROACH

Use a queue. Mark a vertex visited when enqueued, then visit its unvisited neighbors. Store distance[neighbor] = distance[current] + 1.

EXAMPLE

Edges: 0-1, 0-2, 1-3, 2-3. BFS from 0 gives distance 0 to vertex 0, 1 to vertices 1 and 2, and 2 to vertex 3.

COMPLEXITY

With adjacency lists, time is O(V+E) and auxiliary space is O(V).

IMPORTANT

BFS finds shortest paths by edge count in unweighted graphs. For non-uniform non-negative weights, use Dijkstra instead.